

Lecture-12

Mathematics 1 (15B11MA111)

CO [C105.3]

Module: Double Integrals

Topic: Gamma and Beta Functions

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topics to be covered

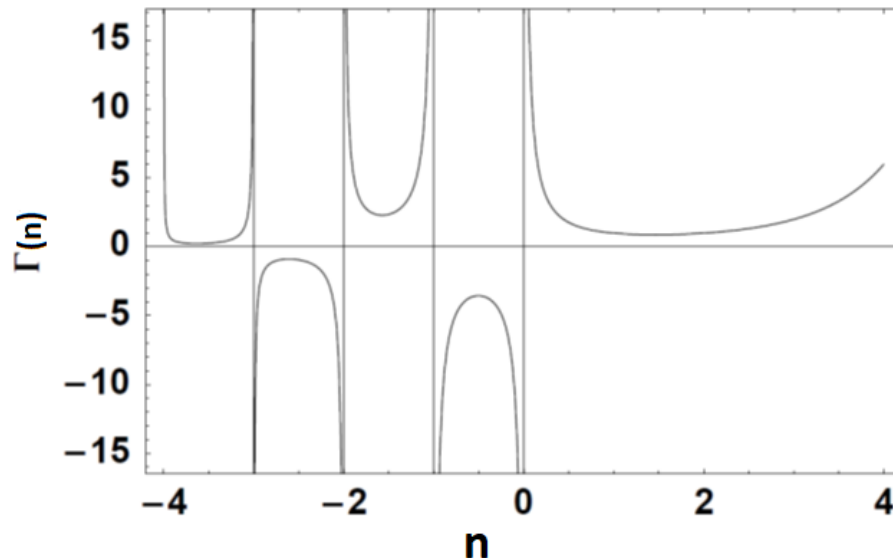
-  **Gamma Function**
-  **Properties of Gamma Function**
-  **Transformations of Gamma Function**
-  **Beta Function**
-  **Transformations of Beta Function**
-  **Some Important Results**
-  **Solved Examples**
-  **Practice Questions**

GAMMA FUNCTION

Definition: The definite integral $\int_0^{\infty} e^{-x} x^{n-1} dx$, for $n > 0$ where n is real and positive is known as the **gamma function** and is denoted by $\Gamma(n)$ (read as ‘Gamma n’).

i.e. $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$, for $n > 0$ (1)

Gamma Function is also called the Eulerian Integral of second kind.



Plot of Gamma Function

Properties of Gamma Function:

1. $\Gamma(1) = 1$.

Proof: We know that $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$, for $n > 0$

Using $n = 1$ in equation of gamma function, we get

$$\Gamma(1) = \int_0^{\infty} e^{-x} x^{1-1} dx = \int_0^{\infty} e^{-x} dx = 1$$

Thus, $\Gamma(1) = 1$.

Properties of Gamma Function (Continued):

2. $\Gamma(n+1) = n \Gamma(n)$.

Proof: We know that, $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$, for $n > 0 \dots (2)$

Replacing n by $n+1$ in (2), we have

$$\Gamma(n+1) = \int_0^\infty e^{-x} x^n dx, \text{ for } n > 0$$

On integrating by parts, we have

$$\begin{aligned}\Gamma(n+1) &= [x^n \cdot (-e^{-x})]_0^\infty - \int_0^\infty (nx^{n-1}(-e^{-x}))dx \\ &= -\lim_{x \rightarrow \infty} x^n \cdot (-e^{-x}) + 0 + n \int_0^\infty x^{n-1} e^{-x} dx \quad (\text{as } \lim_{x \rightarrow 0} x^n e^{-x} = 0, n > 0) \\ &= 0 + 0 + n \int_0^\infty x^{n-1} e^{-x} dx \quad (\text{as } \lim_{x \rightarrow \infty} x^n e^{-x} = 0) \\ &= n \Gamma(n) \quad (\text{using definition of gamma function})\end{aligned}$$

3. If n is a positive integer, then $\Gamma(n+1) = n!$

4. $\Gamma(n) = 0$ if n is 0 or a negative integer.

5. $\Gamma(1/2) = \sqrt{\pi}$.

Transformations of Gamma Function

1. $\Gamma(n) = k^n \int_0^\infty e^{-kx} x^{n-1} dx$, for $n > 0$

Proof: We know that, $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$, for $n > 0$

Put $x = ky$ and $dx = k dy$, we have

$$\begin{aligned}\Gamma(n) &= \int_0^\infty e^{-ky} (ky)^{n-1} (k dy) \\ &= k^n \int_0^\infty e^{-ky} y^{n-1} dy\end{aligned}$$

Or, $\frac{\Gamma(n)}{k^n} = \int_0^\infty e^{-ky} y^{n-1} dy$

2. $\Gamma(n) = \int_0^1 \left(\log \left(\frac{1}{x} \right) \right)^{n-1} dx$, for $n > 0$

Proof: We know that, $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$, for $n > 0$

Put $e^{-x} = y \Rightarrow -x = \log y \Rightarrow x = \log(1/y)$ and $dx = -\frac{dy}{y}$. We get,

$$\begin{aligned}\Gamma(n) &= -\int_1^0 \left(\log \frac{1}{y} \right)^{n-1} dy = \int_0^1 \left(\log \frac{1}{y} \right)^{n-1} dy \\ &= \int_0^1 \left(\log \frac{1}{x} \right)^{n-1} dx\end{aligned}$$

BETA FUNCTION

Definition: The definite integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m > 0$, $n > 0$ is known as the **Beta function** and is denoted by $B(m, n)$ (read as ‘Beta m, n’).

$$\text{i.e. } B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, m > 0, n > 0 \quad (3)$$

Beta Function is also called the Eulerian Integral of first kind.

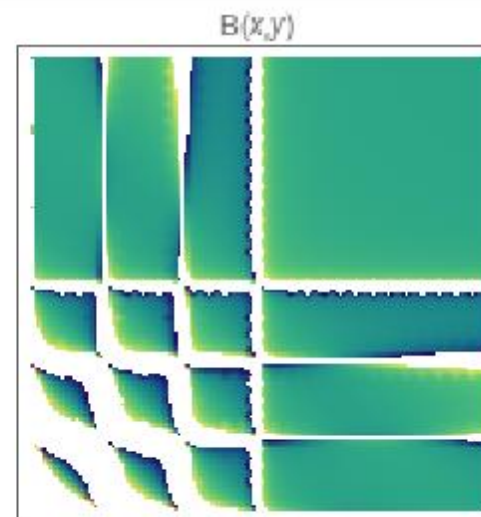
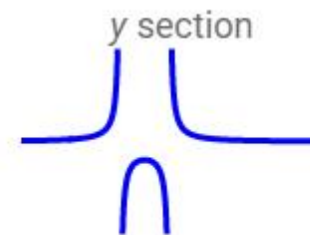
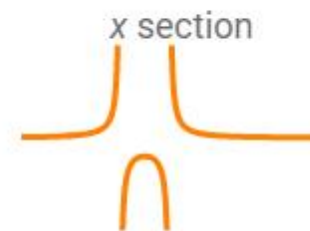
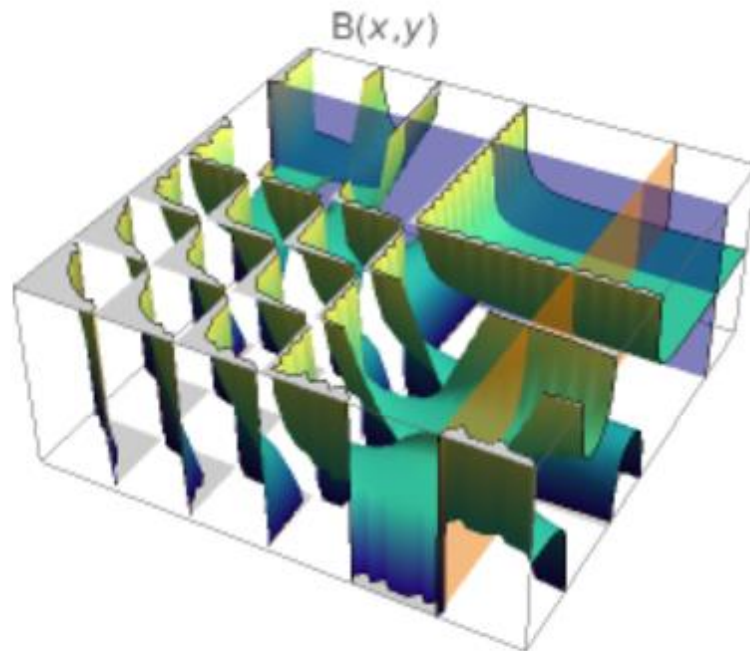
Symmetry of Beta function i.e. $B(m, n) = B(n, m)$

Proof: We know that $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

Using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ in above equation, we get

$$\begin{aligned} B(m, n) &= \int_0^1 (1-x)^{m-1} (1-1+x)^{n-1} dx \\ &= \int_0^1 (x)^{n-1} (1-x)^{m-1} dx = B(n, m) \end{aligned}$$

PLOT OF BETA FUNCTION



Transformations of Beta Function

1. $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

Proof: We know that $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

Put $x = \sin^2 \theta$ and $dx = 2 \sin \theta \cos \theta d\theta$, and

when $x = 0, \theta = 0$ and $\theta = \frac{\pi}{2}$ when $x = 1$.

$$\begin{aligned} B(m, n) &= 2 \int_0^{\pi/2} \sin^{2m-2} \theta (1 - \sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \end{aligned}$$

$$2. \mathbf{B(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx}$$

Proof: We know that $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

$$\text{Put } x = \frac{1}{(1+y)}, \quad dx = -\frac{dy}{(1+y)^2}$$

$$B(m, n) = \int_{\infty}^0 \left(\frac{1}{(1+y)}\right)^{m-1} \left(1 - \frac{1}{(1+y)}\right)^{n-1} \left(-\frac{dy}{(1+y)^2}\right)$$

$$B(m, n) = \int_{\infty}^0 \left(\frac{1}{(1+y)}\right)^{m-1} \left(1 - \frac{1}{(1+y)}\right)^{n-1} \left(-\frac{dy}{(1+y)^2}\right)$$

$$= \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+1}} \left(\frac{1}{(1+y)}\right)^{n-1} dy$$

$$= \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy$$

$$\text{or, } B(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

Some Important Results:

1. Relation between Beta and Gamma Functions

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \quad (4)$$

2. Relation between trigonometric integrals and Gamma function

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)} \quad (5)$$

Remark: For $m = 0$ and $n = 0$ in (5), we have

$$\int_0^{\pi/2} \sin^0 \theta \cos^0 \theta d\theta = \frac{\Gamma\left(\frac{0+1}{2}\right) \Gamma\left(\frac{0+1}{2}\right)}{2 \Gamma\left(\frac{0+0+2}{2}\right)}$$

$$\int_0^{\pi/2} d\theta = [\theta]_0^{\pi/2} = \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{2 \Gamma(1)}$$

$$= \pi = \left(\Gamma\left(\frac{1}{2}\right) \right)^2 \Rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Example: Using Beta and Gamma functions, evaluate:

$$(i) \int_0^{\infty} x^{1/4} e^{-\sqrt{x}} dx \quad (ii) \int_0^1 \left(\frac{x^3}{1-x^3} \right)^{1/2} dx$$

$$(iii) \int_0^1 x^5 (1-x^3)^{10} dx \quad (iv) \int_0^{\infty} \frac{x^8 (1-x^6)}{(1+x)^{24}} dx$$

Solution: (i) Put $\sqrt{x} = t$, $dx = 2t dt$, we have

$$\begin{aligned} \int_0^{\infty} x^{1/4} e^{-\sqrt{x}} dx &= \int_0^{\infty} (t^2)^{1/4} e^{-t} 2t dt \\ &= 2 \int_0^{\infty} t^{3/2} e^{-t} dt \\ &= 2 \Gamma(5/2) = \frac{3}{2} \sqrt{\pi}. \end{aligned}$$

$$(ii) \int_0^1 \left(\frac{x^3}{1-x^3} \right)^{1/2} dx = \int_0^1 x^{3/2} (1-x^3)^{-1/2} dx$$

$$\text{Put } x^3 = y, \quad x = y^{1/3}, \quad dx = \frac{1}{3} y^{-2/3} dy$$

We get,

$$\begin{aligned} &= \int_0^1 y^{1/2} (1-y)^{-1/2} \frac{1}{3} y^{-2/3} dy \\ &= \frac{1}{3} \int_0^1 y^{-1/6} (1-y)^{-1/2} dy \\ &= \frac{1}{3} \int_0^1 y^{\left(\frac{5}{6}\right)-1} (1-y)^{\left(\frac{1}{2}\right)-1} dy = \frac{1}{3} B\left(\frac{5}{6}, \frac{1}{2}\right) \end{aligned}$$

(iii) Put $x^3 = y$, $x = y^{1/3}$, $dx = \frac{1}{3} y^{-2/3} dy$

We get,

$$\begin{aligned} &= \int_0^1 y^{5/3} (1-y)^{10} \frac{1}{3} y^{-2/3} dy \\ &= \frac{1}{3} \int_0^1 y(1-y)^{10} dy \\ &= \frac{1}{3} \int_0^1 y^{2-1} (1-y)^{11-1} dy = \frac{1}{3} B(2, 11) \end{aligned}$$

$$(iv) \int_0^{\infty} \frac{x^8(1-x^6)}{(1+x)^{24}} dx = \int_0^{\infty} \frac{x^8}{(1+x)^{24}} dx - \int_0^{\infty} \frac{x^{14}}{(1+x)^{24}} dx$$

Using $B(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$, we get

$$= \int_0^{\infty} \frac{x^{9-1}}{(1+x)^{9+15}} dx - \int_0^{\infty} \frac{x^{15-1}}{(1+x)^{15+9}} dx$$

$$= B(9, 15) - B(15, 9)$$

$$= 0 \quad (\text{using symmetry of beta function})$$

Example: Show that $\Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi}$

Solution: We know that, $B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$, and

$$B(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\text{Therefore, } \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

Taking $m+n=1$ and $m=1-n$, we get

$$\text{Therefore, } \int_0^{\infty} \frac{x^{n-1}}{(1+x)^1} dx = \frac{\Gamma(n) \Gamma(1-n)}{\Gamma(1)}, 0 < n < 1$$

$$\text{We know that, } \int_0^{\infty} \frac{x^{n-1}}{(1+x)^1} dx = \frac{\pi}{\sin n\pi} \text{ and } \Gamma(1) = 1.$$

Therefore, we get

$$\Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi}$$

Example: Evaluate the integral $\int_0^{\infty} \frac{dx}{1+x^4}$.

Proof: Put $x^4 = y$, $x = y^{1/4}$, $dx = \frac{1}{4} y^{-3/4} dy$

$$\begin{aligned} \text{Therefore, } \int_0^{\infty} \frac{dx}{1+x^4} &= \frac{1}{4} \int_0^{\infty} \frac{y^{-3/4} dy}{1+y} \\ &= \frac{1}{4} \int_0^{\infty} \frac{y^{(\frac{1}{4}-1)} dy}{(1+y)} \end{aligned}$$

We know that, $\int_0^{\infty} \frac{x^{n-1}}{(1+x)^1} dx = \frac{\pi}{\sin n\pi}$. Therefore, we get

$$= \frac{1}{4} \frac{\pi}{\sin \frac{\pi}{4}} = \frac{\pi}{2\sqrt{2}}$$

Example: Prove that $\int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{\pi\sqrt{2}}{2}.$

Proof:
$$\begin{aligned} \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta &= \int_0^{\pi/2} \left(\frac{\sin \theta}{\cos \theta}\right)^{1/2} d\theta \\ &= \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta \, d\theta \end{aligned}$$

We know that,

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta \, d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Therefore, we get
$$\begin{aligned} \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta \, d\theta &= \frac{\Gamma\left(\frac{(1/2)+1}{2}\right) \Gamma\left(\frac{(-1/2)+1}{2}\right)}{2 \Gamma\left(\frac{(1/2)+(-1/2)+2}{2}\right)} \\ &= \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{2 \Gamma(1)} = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right) = \frac{1}{2} \frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}} \end{aligned}$$

Practice Questions:

1. Evaluate $\int_0^1 \frac{dx}{\sqrt{(-\log x)}}$.

Ans. $\sqrt{\pi}$

2. Evaluate the integral $\int_0^1 \frac{x^2 dx}{\sqrt{(1-x^5)}}$ by expressing them in terms of

Beta function.

Ans. $\frac{1}{5} B\left(\frac{3}{5}, \frac{1}{2}\right)$

3. Prove that $\int_0^{\pi/2} \tan^n x \, dx = \frac{\pi}{2} \sec \frac{n\pi}{2}, \quad -1 < n < 1.$

4. Evaluate $\int_{-1}^1 \left(\frac{1+x}{1-x}\right)^{1/2} dx.$

Ans. π

THANK YOU